

Multi – Level linked list

–Delete At Beginning

–Program Analysis

Program:

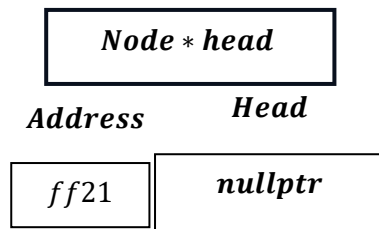
```
void deleteAtBeginning()
{
    if (head == nullptr)
    {
        cout << "List is empty." << endl;
        return;
    }
    Node *temp = head;
    head = head->next;
    // Note: In a full implementation, we should also free
temp->child recursively
    free(temp);
    cout << "Deleted from beginning." << endl;
}
int main()
{
    int choice;
    while (true)
    {
        cout << "5. Delete at Beginning (Main)" << endl;
        cin >> choice;

        switch (choice)
        {
            case 5:
                deleteAtBeginning();
                break;
            default:
                cout << "Invalid choice" << endl;
        }
    }
    return 0;
}
```

Program Analysis

A. When List is Empty

```
if (head == nullptr)
{
    cout << "List is empty." << endl;
    return;
}
```

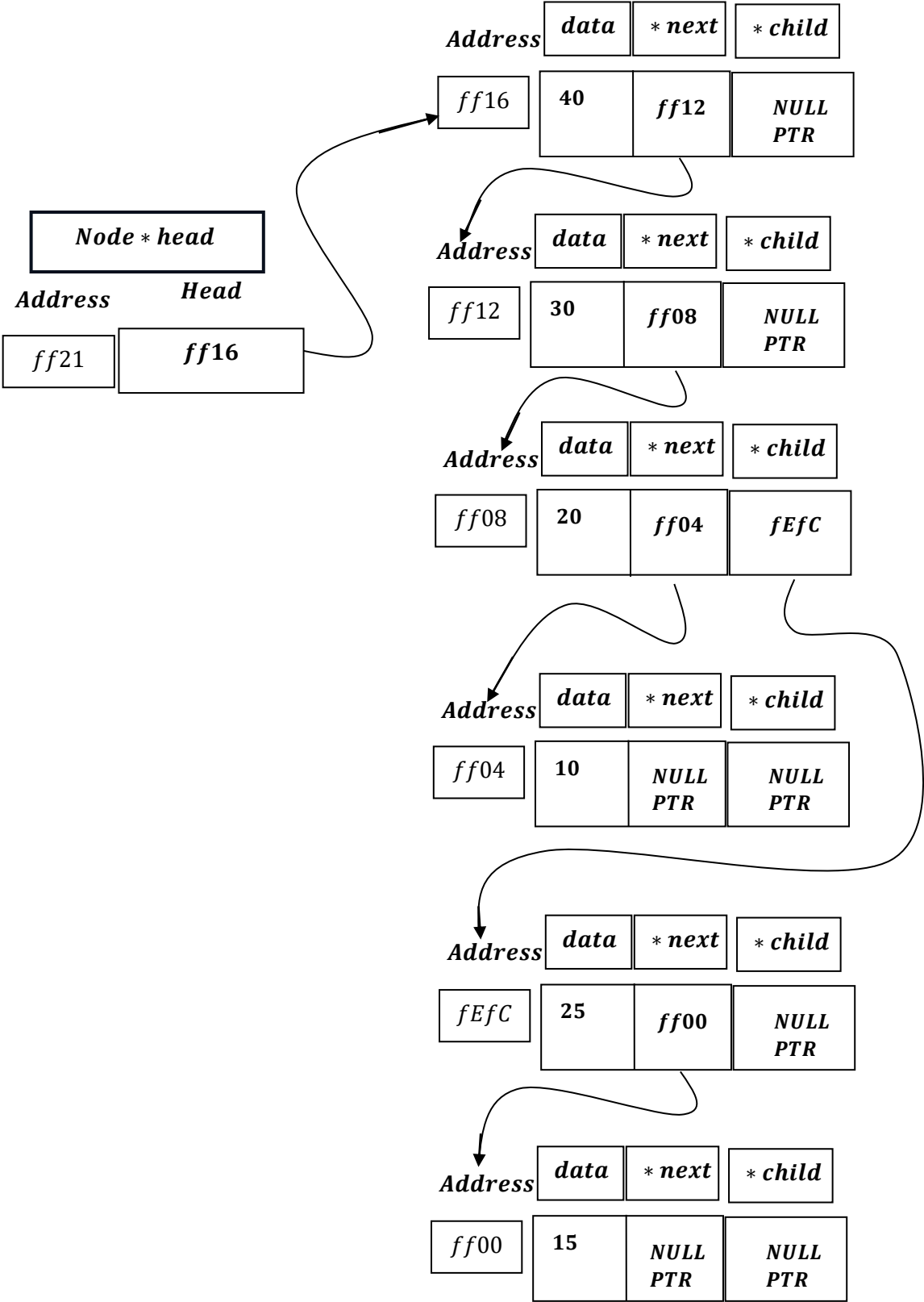


When head = nullptr, means ``List is Empty.``

Hence , prints : ``List is Empty.``

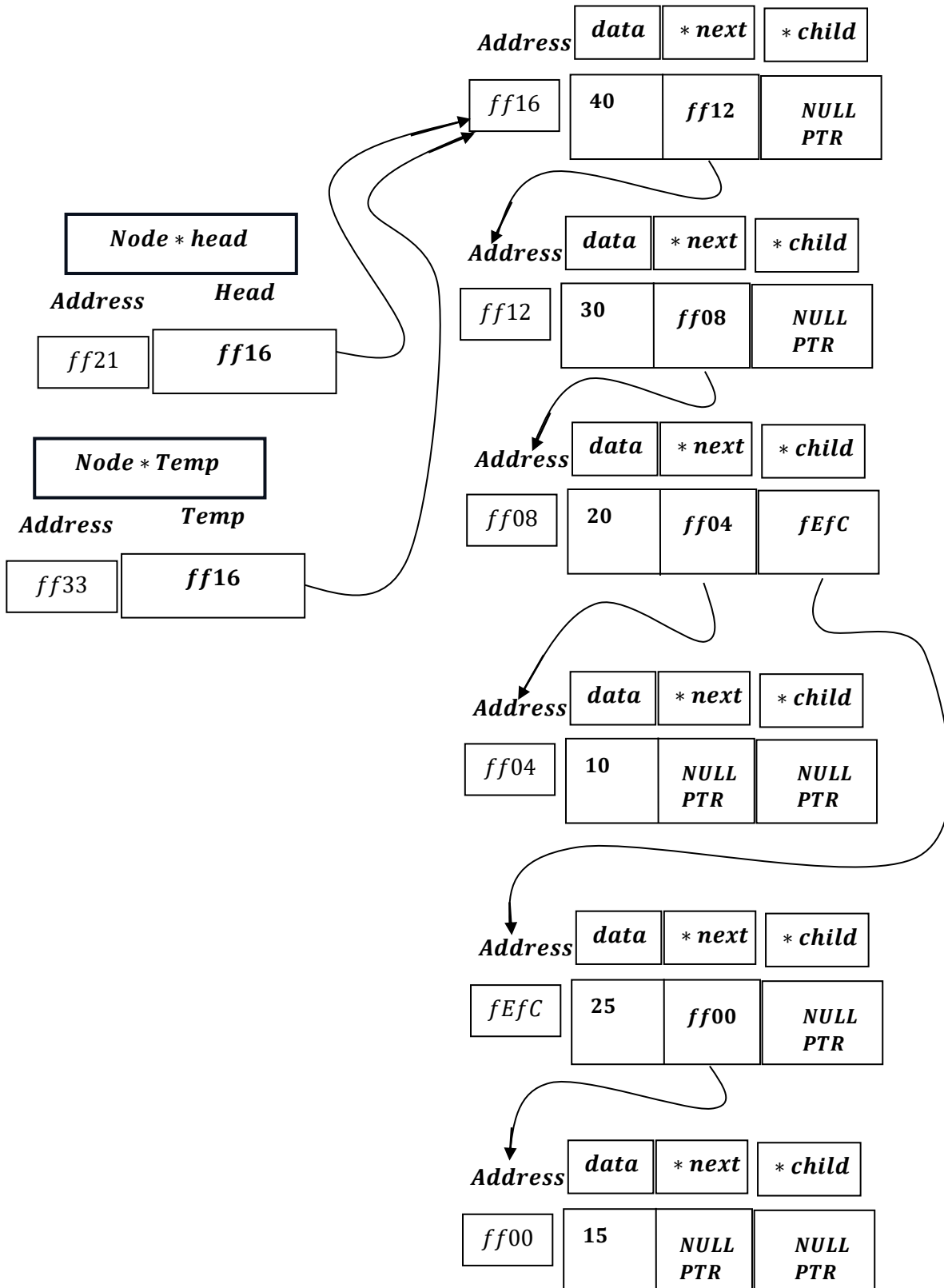
``Return void`` and exit from the function.

B. When List is not Empty



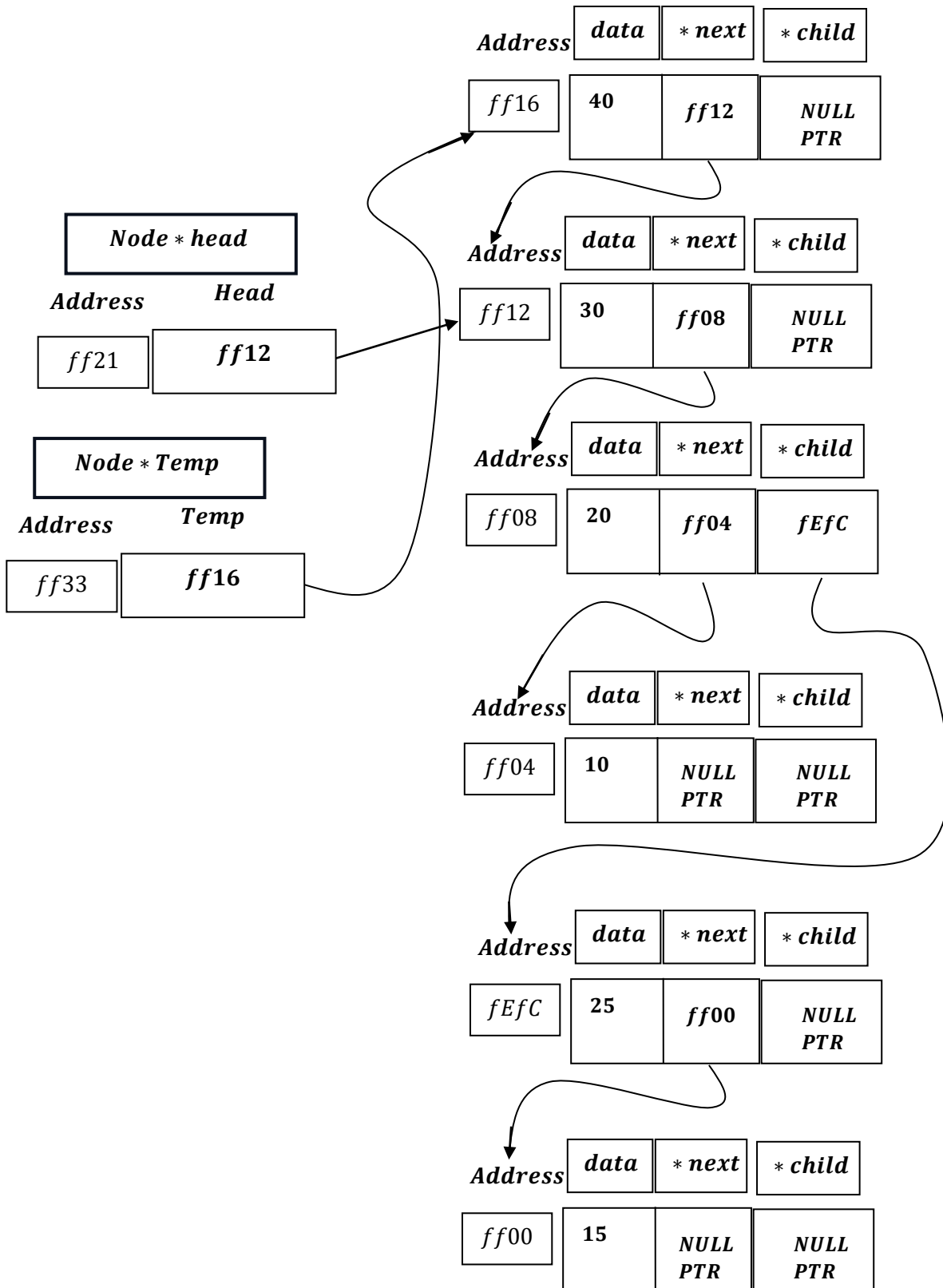
```
Node *temp = head;
```

*Node *Temp = Head = ff16.*

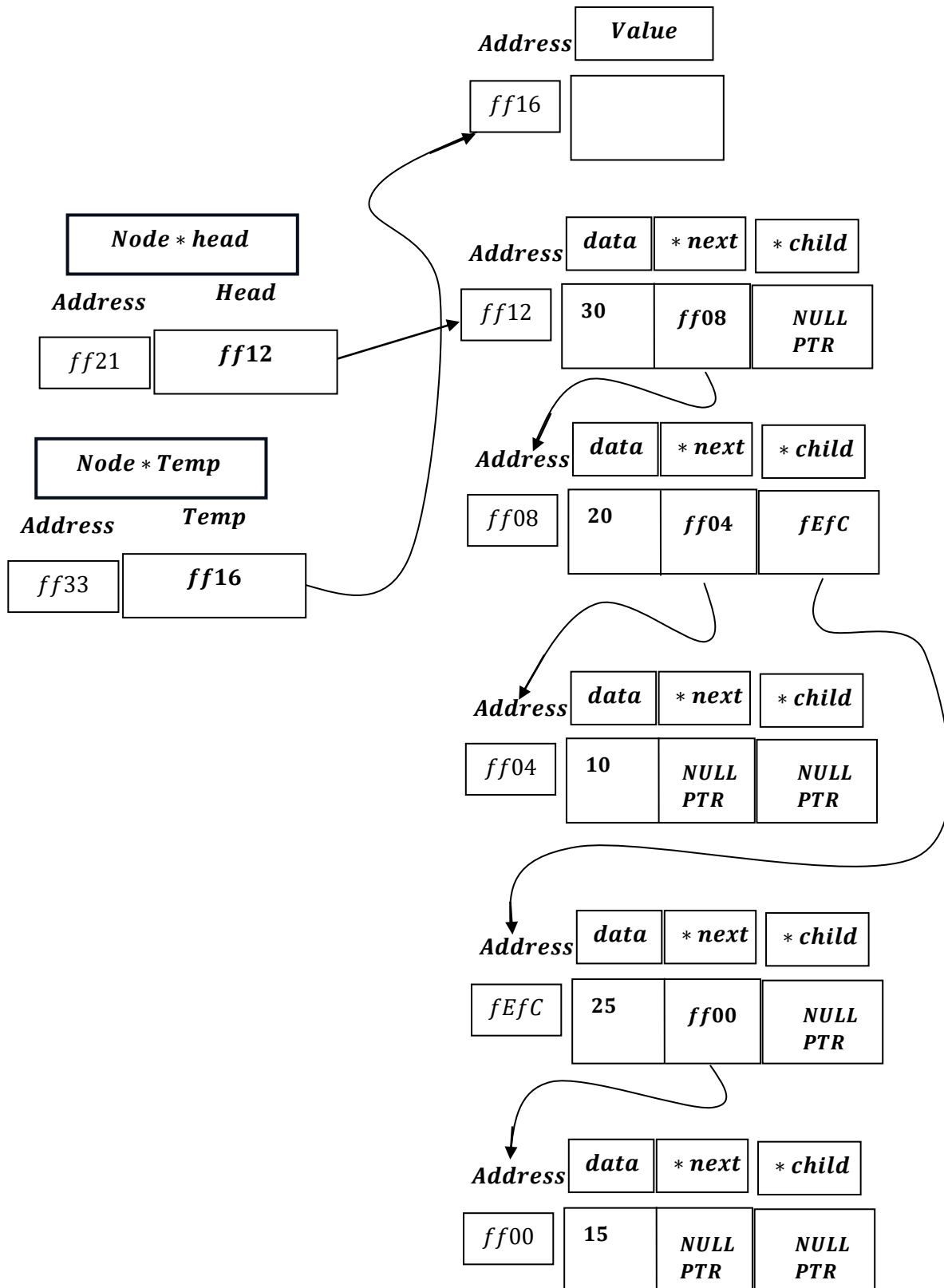


```
head = head->next;
```

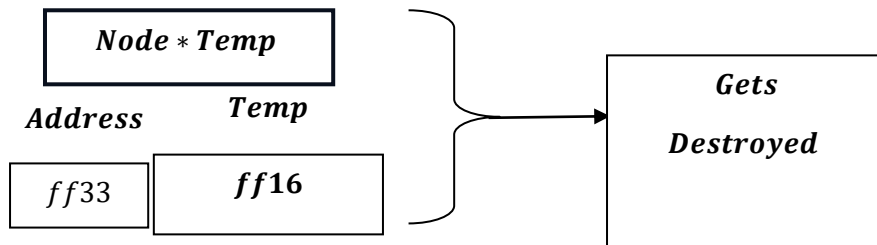
Head = head → next = ff12.



```
free(temp);
```

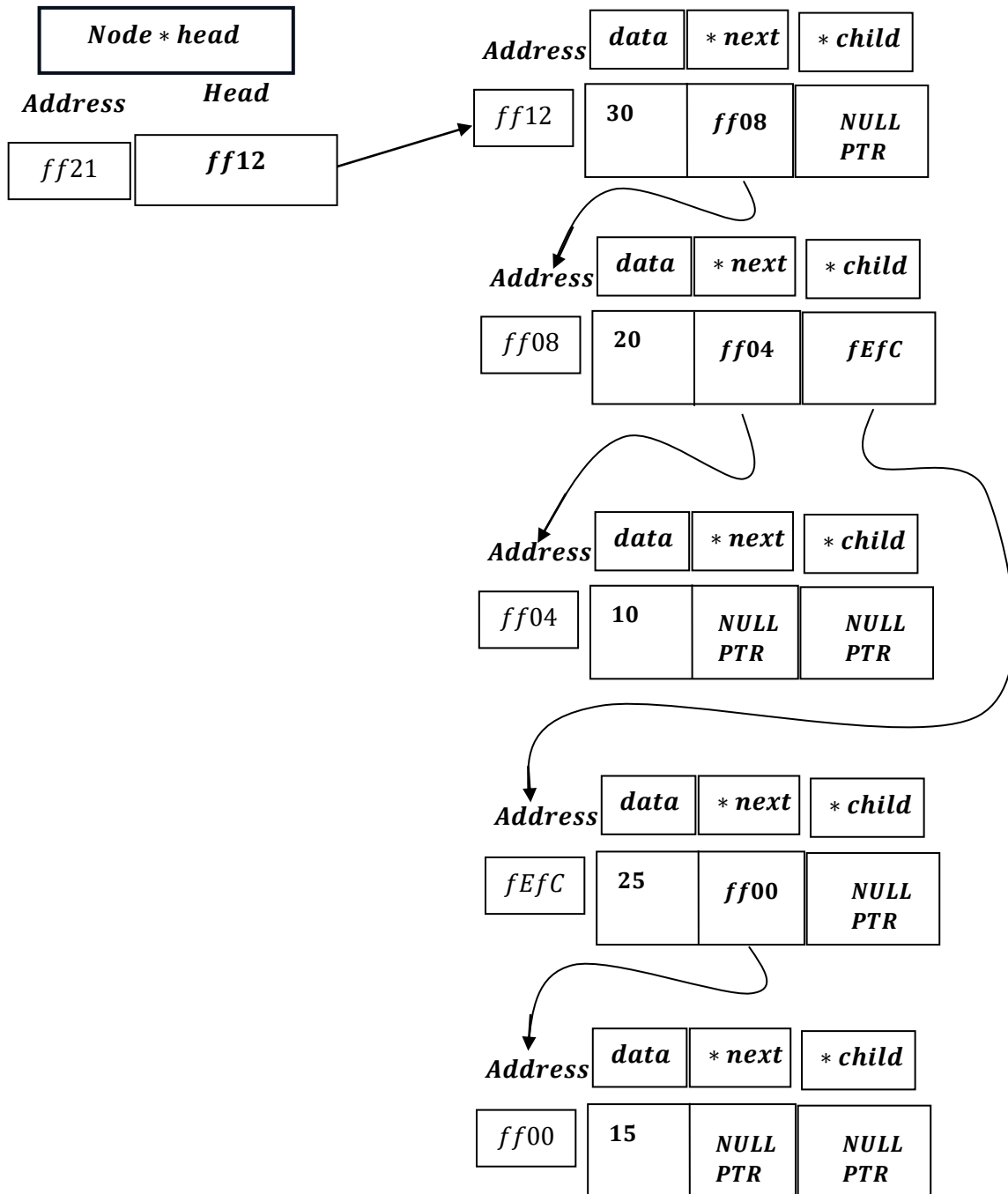


After, DeleteAtBeginning() , function gets over:



As, Temp is local variable of function, DeleteAtBeginning().

And we are left with



Note: Though it delete first node of the Main List , but fails to delete,

1. Child of the main list.

2. Child's child nodes.

3. All deeper levels.

"This deletion function is incomplete for a true multilevel linked list."

Because currently your code only does: free(temp) which frees only: the main node itself, but not the child nodes connected through: temp → child.

Time Complexity

Best Case: When ``List is Empty`` → $\Omega(1)$.

Worst Case: When deleting the first Node , also takes constant time : $O(1)$.

There is no actual average case , hence :

Worst Case = Average Case = $\Theta(1)$.

Hence, $\Omega(1) = O(1) = \Theta(1)$.
